

Cryptographically Secure Audio Watermarking

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1 Introduction

Zero-knowledge proofs (ZKPs), since their seminal introduction in 1985 by Goldwasser, Micali, and Rackoff [GMR85], have fundamentally transformed the landscape of cryptographic protocol design. ZKPs enable a prover to convince a verifier of possessing a witness w such that $F(x; w) = y$, without revealing w . This remarkable property balances privacy and verifiability, enabling secure information exchange while keeping sensitive data confidential.

Traditionally, cryptography has focused primarily on protecting data confidentiality and integrity. By contrast, ZKPs shift the emphasis toward verifiable computation: they allow one to prove that a function was evaluated correctly without disclosing the underlying inputs. A critical challenge in advancing this domain is optimizing the underlying functions evaluated within these proof systems. Therefore, this research proposal focuses on the efficient zero-knowledge proof of computations involving symmetric-key cryptographic primitives, such as hash functions, block ciphers, and pseudorandom functions (PRFs).

1.1 Motivation and Application Landscape

The far-reaching implications of ZKPs have led to their widespread adoption across both academia and industry. Their application domains are diverse and rapidly evolving, including blockchain privacy, decentralized ledger scalability, verifiable storage, and cross-chain interoperability. In the financial and Web3 sectors, ZKPs are central to projects such as Zcash [zca] and Aztec [azt], which use zk-SNARKs to enable private blockchain transactions. Similarly, platforms like StarkWare [sta] and zkSync [zks] employ ZKPs to provide highly scalable layer-2 solutions on Ethereum [eth], while Avail [ava] utilizes them to guarantee verifiable data availability. Beyond blockchain technology, ZKPs are increasingly recognized as critical components in electronic voting (Helios [hel]), privacy-preserving authentication (Microsoft Azure [azu], Polygon ID [Pol], Google Wallet [Goo]), secure delegation of computation to untrusted servers, and privacy-enhanced machine learning.

1.2 Problem Statement

Despite these rapid advances, significant gaps remain between the theoretical potential of ZKPs and their practical deployment. This is especially true where efficient proofs of computation involve symmetric primitives. For example, any ZKP that relies on Merkle proofs necessitates building complex hash function circuits within the proof system. This requirement is the primary bottleneck for rollup-based blockchain scalability solutions, which must constantly prove the correctness of Merkleized state updates. However, mainstream primitives like AES and SHA-3 were not designed for such proof systems. Their internal structures lead to highly inefficient circuit representations, resulting in severe performance overheads. Because ZKP systems prioritize algebraic simplicity over traditional software or hardware efficiency metrics, there is a rapidly growing demand for novel, “ZK-friendly” symmetric-key primitives. Yet, designing such primitives are inherently risky, as their simplified algebraic structures often expose them to algebraic cryptanalysis.

1.3 Research Objectives

This research project aims to resolve the fundamental tension between security and efficiency in symmetric-key primitives designed for advanced cryptographic protocols. The core objective is to systematically investigate, design, and formalize secure primitives tailored for ZKP systems. This will be achieved through a multi-pronged approach:

- O1. Cryptanalysis:** Rigorously evaluate existing ZK-friendly designs (e.g., MiMC, Poseidon, Polocolo) using advanced algebraic methods, including Gröbner basis, resultant-based attacks, and novel FFT-based attacks, to analyze their security weaknesses.
- O2. Formalization:** Develop a comprehensive theoretical framework and robust design guidelines for “ZK-friendly” primitives, ensuring their security and compatibility across diverse ZKP systems (e.g., R1CS, AIRs).

Summary

This project aims to develop symmetric-key primitives that are both highly efficient for advanced cryptographic protocols, such as Zero-Knowledge Proofs, and resilient against cryptanalytic attacks. The Center for Cryptography and Cybersecurity (CCS) at the IIT Hyderabad offers an outstanding environment for advancing research in these fields. My research plan is closely aligned with the center’s core areas, including symmetric-key cryptography and cryptanalysis, and ZKPs for blockchains. My expertise in the design and analysis of symmetric-key primitives is well-suited to investigate these challenges.

2 Research Approach & Feasibility

To ensure systematic progress, the research is organized into two progressive thrusts. The project advances from short-term objectives that build upon established cryptanalytic techniques, toward long-term goals that require fundamentally new theoretical frameworks.

2.1 Thrust 1 (Short-term): Cryptanalysis of Existing ZK-friendly Primitives

Recent years have seen a surge in the proposal of new primitives tailored for ZKPs, such as MiMC [AGR⁺16], the Marvellous family [AAB⁺20], Poseidon [GKR⁺21] and Polocolo [HHL⁺25], to name a few. The trend towards algebraic simplicity in those design significantly alters the cryptanalytic landscape. Classical statistical attacks, such as differential and linear cryptanalysis, become less effective due to the flexibility of field sizes and rapid mixing with few rounds. Instead, as the algebraic representation of these ciphers becomes more concise, their susceptibility to algebraic attacks such as interpolation, GCD attacks, and, most critically, Gröbner basis methods grows. Notably, features that aid efficient proof generation (concise polynomial systems) simultaneously assist algebraic attack algorithms, creating a fundamental tension in balancing efficiency and security. This has led to many proposed constructions later failing to achieve their claimed security, highlighting the need for more extensive cryptanalysis in this area. Currently, I am investigating the Gröbner basis attacks on the Polocolo hash function [HHL⁺25], designed for R1CS and PLONK-based proof systems. I am also exploring the possibility of applying FFT-based key recovery attacks that I have proposed in [DGK⁺24] for AES, on the ciphers defined over prime field ($\mathbb{F}_p^n$) such as AES-prime [MMMS23], which was proposed for efficient use in MPC protocols.

2.2 Thrust 2 (Long-term): Formalization of Design Criteria for ZK-friendly Primitives

The practical realization of ZKPs advanced significantly with the advent of zero-knowledge succinct non-interactive arguments of knowledge (zk-SNARKs), first introduced by Ben-Sasson et al. [BCTV14] in 2013. Since then, the field has rapidly expanded to incorporate diverse zk-SNARK protocols, such as Pinocchio [PHGR16], Groth16 [Gro16], PLONK [GWC19], and many others, as well as Zero-Knowledge Scalable Transparent Arguments of Knowledge (zk-STARKs) [BBHR19]. Notably, different ZKP frameworks use distinct algebraic models depending on their proof system and efficiency goals. For example, zk-SNARKs typically employ Rank-1 Constraint Systems (R1CS) with Quadratic Arithmetic Programs (QAPs), while zk-STARKs utilize Algebraic Intermediate Representations (AIRs). Each framework presents unique trade-offs in proof size, verification speed, trust assumptions, and compatibility with various types of symmetric-key primitives. My long-term objective is to develop a comprehensive formalization of the fundamental principles underpinning these frameworks and to articulate robust design guidelines for arithmetization-friendly primitives. By deepening the theoretical foundations of algebraic cryptanalysis, including refining

existing methods such as Gröbner basis attacks, resultant computations, I aim to enable both rigorous security evaluation and efficient design.

3 Expected Deliverables

To ensure measurable progress, the concrete deliverables for this project are outlined below by research thrust. The anticipated schedule mapping these tasks and milestones across the 24-month project lifecycle is presented in Figure 1.

Thrust 1 – Cryptanalysis of Existing Ciphers

D1. Cryptanalytic results on deployed ciphers. Published attacks in terms of finding new distinguishers and key-recovery results on at least two of the target ciphers, targeting publication at CRYPTO, EUROCRYPT, ASIACRYPT, or IACR ToSC.

D2. High performance cryptanalysis using GPU. Cryptanalysis of ZK-friendly block ciphers often required solving system of non-linear equations over large finite fields. We will develop GPU-accelerated solvers for these equations, enabling fast and efficient cryptanalysis of ZK-friendly ciphers. An open-source library will be made available for reproducible and extensible cryptanalysis by the wider community.

Thrust 2 – Formalization of Design Criteria

D5. Design Guidelines for ZK-Friendly Primitives. Building on the cryptanalytic insights gained from Thrust 1, I aim to formalize the design criteria needed to balance efficiency and security across different ZKP frameworks. This is intended to serve as a foundational guide, helping the broader cryptographic community better navigate these complex design trade-offs.

D6. Theoretical Advancements in Algebraic Cryptanalysis. Publication of refined theoretical methods for algebraic cryptanalysis, specifically improving the efficiency and applicability of Gröbner basis and resultant-based attacks on algebraically simple primitives.

Training and Capacity Building

D7. Human resource development. Rigorous training and mentorship of one Ph.D. scholar and two M.Tech. students at the highly specialized intersection of symmetric-key cryptography, advanced protocols, and automated cryptanalysis. Graduates equipped with these skill sets are currently in high demand across both academia and the industry.

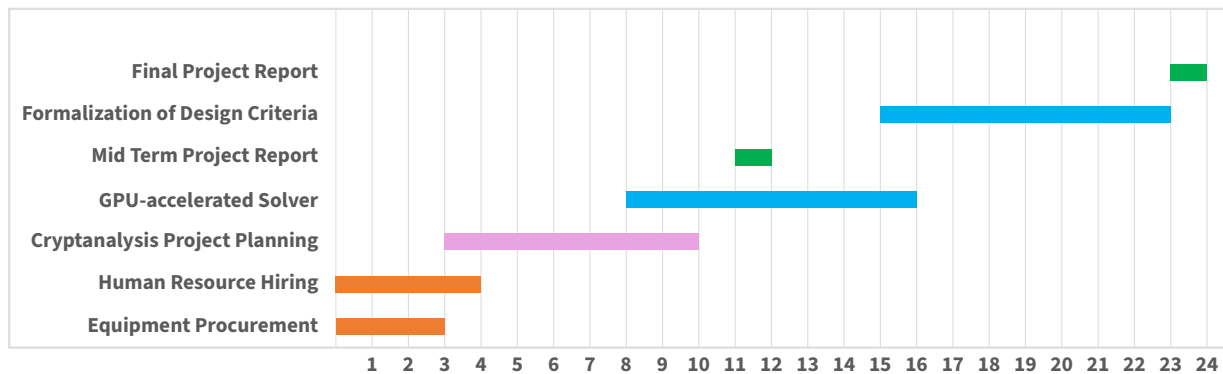


Figure 1: 24 Months Timeline

4 Budget Allocation

The proposed budget of **Rs. 35.86 Lakhs** over a period of two years is carefully structured to meet the project's specific manpower, equipment, and dissemination requirements. Heavy computational infrastructure, such as multi-GPU servers for running SAT/MILP solvers and algebraic

cryptanalysis, is either already available or can be provisioned on-demand through the **Center for Cryptography and Cybersecurity (CCS)** at **IIT Hyderabad**. Consequently, the requested equipment budget focuses exclusively on high-performance portable client machines necessary for the research team to conduct algorithm design, code development, debugging, and daily analytical tasks. The detailed financial breakdown is provided in Table 1.

Category	Description	Amount (Rs.)
Equipment	High-Performance Portable Workstation: 1 × Apple MacBook Pro (M5 Pro/Max chip) with 32GB+ unified memory. Required for advanced prototyping, local code compilation, and running localized solver benchmarks.	3,90,800
	GPU-Accelerated Portable Workstation: 1 × High-Performance AMD Laptop (e.g., ASUS ROG Strix SCAR 17 or Lenovo Legion Pro, featuring AMD Ryzen 9 7945HX, 32GB RAM, and NVIDIA RTX 4080/4090 GPU). Assigned to the research scholars for GPU-accelerated cryptanalysis, heavy solver execution, and algorithm design.	3,00,000
	<i>Equipment Subtotal</i>	<i>6,90,800</i>
Manpower	Ph.D. Scholar Fellowship (1 Student): Support for one Ph.D. student for up to 3 years (Rs. 37,000/month + HRA for the first two years and Rs. 42,000/month + HRA for the last year).	11,16,000
	M.Tech Student Fellowships (2 Students): Support for two M.Tech students for 3 years (Rs. 15,000/month/student).	10,80,000
	<i>Manpower Subtotal</i>	<i>21,96,000</i>
Travel	Conference presentations, networking, and collaborative visits to disseminate research outcomes at top-tier venues.	4,00,000
Contingency	Cloud compute credits (AWS/GCP), software licenses, system maintenance, minor electronic components, and lab setup.	2,13,200
Total		35,00,000

Table 1: Detailed Breakdown of the Proposed Budget

5 Conclusion

This project is dedicated to the design and analysis of symmetric primitives tailored for efficient integration with zero-knowledge proof systems. Addressing these specialized requirements presents both opportunities and challenges for cryptography research. Through this research, I aim to advance both the theoretical foundations and the practical deployment of tailored primitives, bridging the gap between robust security guarantees and computational efficiency. The resulting analysis and frameworks will directly support the standardisation community by providing the rigorous cryptanalytic evidence and clear metrics for technical evaluation.

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